# Pramish Thapa

London, UK | +44 7587 402388 | pramish.thapa17@gmail.com
orcid.org/0009-0001-7658-8862 | linkedin.com/in/pramish-thapa-36b828156 | github.com/Pramish17
pramishthapa.com | battery-rul-dashboard.vercel.app

## PROFILE

AI researcher and software engineer. First-author manuscript in preparation for MDPI Sensors on cross-dataset benchmarking of ML degradation models, with emphasis on statistical robustness and uncertainty quantification. MSc Software Engineering with Distinction and nearly three years of production engineering experience. Research interests: trustworthy and reliable AI, security of LLM agents, reproducible ML evaluation, and uncertainty quantification.

## EDUCATION

**University of West London, UK** Sep 2023 - Sep 2024

*MSc Software Engineering, Distinction (awarded 18 October 2024)*

- Dissertation (85/100): digital twin and IoT pipeline for building energy monitoring; Random Forest model achieving $R^2$ of 0.999 on held-out data.

**Crimson College of Technology, Pokhara University, Nepal** Sep 2016 - Jul 2021

*Bachelor of Computer Information Systems (BCIS), CGPA 3.31/4.0*

- Modules graded A include Major Project, Data Structures and Algorithms, and Mobile Computing.

## RESEARCH AND PUBLICATIONS

- First-author manuscript in preparation for MDPI Sensors (with Dr F. Dinmohammadi, University of West London): "No Single Model Dominates: A Regime-Aware, Cross-Dataset Benchmark of Degradation Models for Lithium-Ion Battery Remaining Useful Life". Seven models including CNN-LSTM, Gaussian process regression and XGBoost benchmarked across the NASA PCoE and Oxford datasets under a chronological evaluation protocol; bootstrap confidence intervals, Wilcoxon tests with Holm correction, rolling-origin validation and leave-one-cell-out jackknife.
- Ongoing: prompt-injection robustness of LLM agents. Reproducing the AgentDojo benchmark (ETH Zurich) on current Claude models, measuring task utility and attack success: github.com/Pramish17/agent-reliability
- Interactive research dashboard (React / Recharts) reproducing every figure and metric in the manuscript: battery-rul-dashboard.vercel.app
- CNN-LSTM remaining-useful-life prediction on NASA C-MAPSS turbofan degradation data, RMSE 14.84, with SHAP interpretability analysis.

## TECHNICAL SKILLS

**Machine learning:** Python, PyTorch, TensorFlow, scikit-learn, pandas, NumPy, SHAP; CNN, LSTM and hybrid architectures; Gaussian process regression, XGBoost, Random Forest

**Software engineering:** Java, Spring Boot, REST APIs, Google Cloud Platform, Docker, Jenkins CI/CD, Git, SQL (MySQL, PostgreSQL), JavaScript / React.

## PROFESSIONAL EXPERIENCE

**Independent research and upskilling** Oct 2024 - Present

*Self-directed research and training, London, UK*

**Oncore Corporation (client: Equifax)** Sep 2021 - Jun 2023

*Java Developer, Kathmandu, Nepal*

- Built and maintained Java / Spring Boot microservices on Google Cloud for Equifax credit-risk and identity-verification systems.
- Designed asynchronous data pipelines (Cloud Pub/Sub, Firestore) deployed through Jenkins CI/CD, with 95%+ test coverage on owned services.
- Built services for a regulated financial-services client; presented technical work to client stakeholders in 50+ review meetings.

**InfoxIT Pvt. Ltd** Jan 2020 - May 2021

*Java Intern, Nepal*

- Developed backend ERP modules in Java, Spring and Hibernate for banking clients.
- Designed MySQL schemas and optimised queries based on user feedback.

## FURTHER TRAINING

Neural Networks and Deep Learning, DeepLearning.AI (2026) | Full Stack Developer Traineeship, IT Career Switch (2025 - 2026) | Full-Stack Engineer Career Path, Codecademy (2025)

## REFEREES

**Dr Fateme Dinmohammadi**, Associate Professor in Artificial Intelligence, University of West London | fateme.dinmohammadi@uwl.ac.uk (supervisor and co-author)

**Dr Thomas Madsen**, Lead for Teaching and Quality, University of West London | thomas.madsen@uwl.ac.uk

**Yogya Sharma**, Founder, Oncore Corporation | yogya.sharma@omglobaltech.com